

# PHISHING DETECTION & AWARENESS REPORT

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Future Interns Cybersecurity Task No. 2

Real-World Phishing Email Analysis  
15 Case Studies with Technical Deep-Dive

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## EXECUTIVE SUMMARY

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Phishing remains the most common and effective cyber attack vector. This report analyzes 15 real phishing email samples drawn from public honeypot datasets and labeled phishing collections. Each sample is dissected — headers, links, sender reputation, and content patterns — then classified using a 3-tier risk system: Safe / Suspicious / Phishing.

### KEY FINDINGS

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| Metric                         | Value                                                                   |
|--------------------------------|-------------------------------------------------------------------------|
| Samples analyzed               | 15 (from phishing_pot + Sadat dataset)                                  |
| Classified as Phishing         | 6 (40%)                                                                 |
| Classified as Suspicious       | 5 (33%)                                                                 |
| Classified as Safe             | 4 (27%)                                                                 |
| Unique attack types identified | 6 (credentials, 419 scams, malware, BEC, giveaways, subscription fraud) |

|                                       |                                        |
|---------------------------------------|----------------------------------------|
| Languages represented                 | English, Portuguese, German, Norwegian |
| Active phishing URLs cross-referenced | 57,971                                 |

## TOP INDICATORS FOUND

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- SPF/DKIM/DMARC failures — 8 of 15 samples (53%)
- Urgency/fear language — 5 of 15 samples (33%)
- Domain mismatch (From vs Return-Path) — 10 of 15 samples (67%)
- Generic greetings — 3 of 15 samples (20%)
- Suspicious links — 14 of 15 samples (93%)

# METHODOLOGY & DATA SOURCES

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## Data Sources

| Source                 | Type                              | Samples Used    | Purpose                                           |
|------------------------|-----------------------------------|-----------------|---------------------------------------------------|
| phishing_pot           | Real .eml files with full headers | 15              | Header analysis, body inspection, link extraction |
| Phishing_Email (Sadat) | Labeled CSV (phishing/legitimate) | 163 rows        | Pattern validation, false positive test           |
| Phishing.Database      | 57,971 active phishing URLs       | Cross-reference | Link/domain reputation checking                   |

## Analysis Steps

- Header extraction — Parse Received chain, SPF, DKIM, DMARC, Return-Path, SCL
- Body decoding — Decode base64/quoted-printable bodies, extract text and HTML
- Link enumeration — Extract all URLs, check against Phishing.Database
- Sender verification — Check domain alignment, lookalike detection
- Content analysis — Urgency language, generic greetings, brand impersonation
- Risk scoring — Weighted scoring (auth 40%, links 30%, content 20%, sender 10%)
- Classification — 3-tier: Safe (0-1 pts), Suspicious (2-4 pts), Phishing (5+ pts)

# PHISHING LANDSCAPE OVERVIEW

## Attack Types Identified

| Type                   | Description                                       | Count |
|------------------------|---------------------------------------------------|-------|
| Credential Harvesting  | Fake login pages for banks, cryptocurrency, email | 5     |
| Advance-Fee (419) Scam | Inheritance/fund transfer fraud                   | 1     |
| Brand Impersonation    | Fake alerts from known companies                  | 4     |
| Giveaway/Lottery Scam  | Fake prize notifications                          | 2     |
| Subscription Fraud     | Fake subscription cancellation notices            | 1     |
| Malware Delivery       | Links to malicious downloads                      | 2     |

## Common Tactics

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Fear + urgency: "Your account will be locked", "Unusual activity detected"

Spoofed sender names: Display name matches real brand, but email domain is fake

Lookalike domains: atendimento.com.br instead of bradesco.com.br

URL shorteners & redirects: dsgo.to, zzdzw.com

Mass/BCC recipients: Multiple undisclosed recipients

HTML-only emails: No plain-text alternative (bypasses some filters)

# COMMON PHISHING INDICATORS

## REFERENCE GUIDE

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### Technical Indicators (Email Headers)

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| Indicator              | What to Look For                               | Severity |
|------------------------|------------------------------------------------|----------|
| SPF Fail/None/SoftFail | spf=fail or spf=none in Authentication-Results | High     |
| DKIM Fail/None         | dkim=fail or dkim=none                         | High     |
| DMARC Fail/None        | dmARC=fail or dmARC=none                       | High     |
| Compauth Fail          | compauth=fail in header                        | High     |
| Domain Mismatch        | From domain ≠ Return-Path domain               | Medium   |
| Unknown Sender IP      | IP not associated with claimed organization    | Medium   |
| High SCL               | X-MS-Exchange-Organization-SCL: 5-9            | Medium   |

|                        |                          |     |
|------------------------|--------------------------|-----|
| Missing DKIM Signature | No DKIM-Signature header | Low |
|------------------------|--------------------------|-----|

## Content Indicators (Email Body)

| Indicator               | Example                                              | Severity |
|-------------------------|------------------------------------------------------|----------|
| Urgency/Fear            | "Your account will be locked", "Suspicious activity" | High     |
| Generic Greeting        | "Dear Customer", "Dear User"                         | Medium   |
| Grammar Errors          | "Immediate" (sample-12), odd capitalization          | Medium   |
| Mismatched Branding     | Coca Cola offer from stayfriends.de                  | High     |
| Spoofed Display Name    | "C o i n b a s e" with spaces                        | High     |
| Suspicious Links        | URL shorteners, lookalike domains                    | High     |
| Unsolicited Offers      | "You won!", "400% bonus", "Free"                     | Medium   |
| Request for Credentials | "Verify now", "Login to confirm"                     | Critical |

# RISK CLASSIFICATION FRAMEWORK

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## 3-Tier System

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Score 0-1:  **SAFE** — Deliver normally

Score 2-4:  **SUSPICIOUS** — Quarantine, verify via secondary channel

Score 5+:  **PHISHING** — Block, report, security team notified

## Scoring Weights

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| Category          | Max Points | Indicators                                 |
|-------------------|------------|--------------------------------------------|
| Authentication    | 8          | SPF (2), DKIM (2), DMARC (2), Compauth (2) |
| Content           | 3          | Urgency (2), Generic greeting (1)          |
| Sender            | 2          | Domain mismatch (1), Unknown sender (1)    |
| Links/Attachments | 2          | Suspicious links (1), Attachments (1)      |
| Total             | 15         | —                                          |

# PREVENTION GUIDELINES

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## Do's and Don'ts for All Employees

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DO

| # | Action                                             | Why                                               |
|---|----------------------------------------------------|---------------------------------------------------|
| 1 | Verify sender email address, not just display name | Display names are easily faked                    |
| 2 | Hover over links before clicking                   | Check the real URL matches the displayed one      |
| 3 | Type addresses manually in your browser            | Avoids link redirects                             |
| 4 | Enable Multi-Factor Authentication (MFA)           | Stops credential theft even if password is stolen |
| 5 | Report suspicious emails to IT/security team       | Helps protect others                              |
| 6 | Use separate passwords for every account           | Limits damage if one is compromised               |
| 7 | Keep software updated                              | Patches vulnerabilities attackers exploit         |
| 8 | Trust your gut — if it feels wrong, it probably is | Attackers rely on rushing you                     |



## DON'T

| # | Action                                            | Why                                                      |
|---|---------------------------------------------------|----------------------------------------------------------|
| 1 | Don't click links in unexpected emails            | Even "Unsubscribe" links can be malicious                |
| 2 | Don't open unexpected attachments                 | .zip, .exe, .docm, .js files are common malware carriers |
| 3 | Don't provide personal info via email             | Legitimate companies never ask for passwords via email   |
| 4 | Don't act on urgency                              | "Act now or account locked" is a classic tactic          |
| 5 | Don't reply to suspicious emails                  | Confirms your email is active to spammers                |
| 6 | Don't forward phishing emails to colleagues       | Send to IT instead — they'll analyze safely              |
| 7 | Don't call numbers in suspicious emails           | Call the company's official number instead               |
| 8 | Don't use public Wi-Fi for sensitive transactions | Attackers can intercept traffic                          |

# QUICK PREVENTION TIPS (1-MINUTE READ)

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## How to Spot a Phishing Email in 10 Seconds

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### 1. CHECK THE SENDER

✗ noreply@access-accsecurity.com

✓ noreply@microsoft.com

### 2. HOVER OVER LINKS

✗ <https://zzdzw.com/> (not binance.com)

✓ <https://www.binance.com/en/verify>

### 3. LOOK FOR URGENCY

✗ "Your account will be locked in 24hrs"

✓ "Please review when you have a moment"

### 4. CHECK THE GREETING

✗ "Dear Customer / Dear User"

✓ "Dear [Your Name]"

### 5. TRUST YOUR INSTINCT

If something feels off, it probably is. Report it to IT immediately.

## If You Think You've Been Phished

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- Change your password immediately — on the real website, not from the email link
- Enable MFA if not already active
- Contact IT/Security and report the incident
- Check account activity for unauthorized logins or transactions
- Scan your computer for malware
- Watch for follow-up attacks — attackers often strike again

## CONCLUSION

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Phishing attacks are evolving constantly, with attackers using increasingly sophisticated techniques to bypass technical controls. The best defense is a combination of:

- ✓ Technical controls (SPF, DKIM, DMARC, email filtering)
- ✓ User awareness and training
- ✓ Incident response procedures
- ✓ Regular phishing simulations and security audits

By understanding the tactics, techniques, and indicators of compromise covered in this report, organizations and individuals can significantly reduce their risk of falling victim to phishing attacks.

**Stay vigilant. Stay secure.**